

Project Information

Project Type: Team

Student Name(s): Jaydine Stiles, Joseph Ko

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Research Title: Post-Disaster Hospital Simulation with Digital Twins and Adaptive Task Allocation

Problem Statement, Hypothesis, and Research Questions:

Problem Statement:

In emergency scenarios like post-earthquake hospital rescues, robotic agents must coordinate tasks such as triage, transport, and supply delivery in real time under uncertain, hazardous conditions. These scenarios demand not only rapid decision-making but also resilience to changing environments, incomplete information, and varying levels of agent reliability.

To address these challenges, digital twins—virtual replicas of physical agents and environments—can serve as a central coordination layer by enabling real-time monitoring, planning, and anomaly detection. However, effectively using digital twins for multi-agent task allocation requires optimizing across complex, dynamic constraints such as trust levels, blocked routes, and patient urgency.

This project investigates how quantum-enhanced graph optimization (via QUBO modeling) can improve digital twin-assisted coordination in a simulated rescue setting. The aim is to evaluate whether quantum-inspired methods can outperform classical task allocation strategies when trust, uncertainty, and environmental risk are encoded as graph attributes.

Hypotheses:

Integrating dynamic trust and risk data into a Neo4j knowledge graph and driving task assignment and routing with a quantum-enhanced optimizer will significantly improve rescue performance—reducing average mission completion time and increasing task success rates—compared to a classical graph-based optimization approach under the same disrupted conditions.

Research Question:

How does our Neo4j + quantum-enhanced optimizer compare to a classical Dijkstra/QUBO hybrid in terms of average mission completion time and patient extraction success rate in a simulated post-earthquake hospital wing with dynamic trust scores?

Metrics: mean completion time, % tasks completed, average trust score of assigned agents.

Literature Review

1. Safety Control of Service Robots with LLMs and Embodied Knowledge Graphs
 - Authors: Yong Qi, Gabriel Kyebambo, Siyuan Xie, Wei Shen, Shenghui Wang, Bitao Xie, Bin He, Zhipeng Wang, Shuo Jiang
 - Publication Year: 2024

Summary:

This paper proposes a new safety framework for service robots by combining Large Language Models (LLMs), Embodied Robotic Control Prompts (ERCs), and Embodied Knowledge Graphs (EKGs). The goal is to improve how robots understand commands, break them down into safe task steps, and validate those steps before taking action. The ERCs help LLMs generate clear, safety-focused task plans. The EKG acts like a dynamic fact-checker, making sure every step of the plan is safe and fits the current environment. The system was tested against LLM-only, KG-only, and LLM+KG systems. It achieved a 90% task success rate, fewer errors (3%), and fewer safety violations, showing stronger adaptability in changing environments. The study highlights the benefit of adding real-time validation and feedback loops to robot planning, but it notes challenges in scaling the knowledge graph and ensuring real-time performance in complex environments. This work informs how we might design Neo4j-based graphs that both represent knowledge and help robots act safely in dynamic rescue scenarios.

Relevance to Our Project:

This paper shows how a knowledge graph can do more than store data—it can actively validate plans for safety before they are executed. We can apply this idea by designing a Neo4j graph that checks task assignments and paths for hazards (e.g., blocked corridors) in our hospital rescue scenario, improving safety before sending tasks to the quantum optimizer.

Link: [arXiv:2405.17846](https://arxiv.org/abs/2405.17846)

2. UAV-UGV Cooperative Trajectory Optimization and Task Allocation for Medical Rescue Tasks in Post-Disaster Environments

Authors: Kaiyuan Chen, Wanpeng Zhao, Yongxi Liu, Yuanqing Xia, Wannian Liang, Shuo Wang
Publication Year: 2025

Summary:

This paper presents an integrated framework that coordinates unmanned aerial vehicles (UAVs) and unmanned ground vehicles (UGVs) for medical rescue operations in post-disaster environments. The authors address two main challenges: (1) how to assign tasks efficiently to a mixed fleet of UAVs and UGVs and (2) how to generate safe, efficient, and smooth trajectories through obstacle-filled environments.

Their solution combines an enhanced Genetic Algorithm (GA) for task allocation, Informed-RRT* for initial collision-free path planning, and Covariance Matrix Adaptation Evolution Strategy (CMA-ES) for optimizing task sequences and refining trajectories. This approach leverages the complementary strengths of UAVs (fast, agile) and UGVs (strong, high-capacity) to maximize mission efficiency.

The proposed method was tested through simulations in a realistic, earthquake-damaged urban scenario. Results showed that their system significantly outperformed simpler approaches. Specifically, it reduced mission completion time and total path length compared to:

- Random assignment (tasks given out randomly, no planning)
- K-Means clustering (tasks grouped by location without considering robot type or obstacles)
- Basic GA without advanced path optimization

Their method completed missions in as little as ~27 minutes, compared to ~100 minutes for random and K-Means methods, and showed strong scalability as the number of tasks and vehicles increased. The framework successfully generated collision-free, efficient routes and demonstrated that coordinated UAV-UGV planning can dramatically improve disaster response performance.

Relevance to Our Project:

We could use their random assignment results as a baseline for constructing our own tables of data, helping us compare more advanced trust-aware or quantum-enhanced task allocation strategies against a simple starting point.

Link: <https://arxiv.org/pdf/2506.06136>

3. Robotics Classification of Domain Knowledge Based on a Knowledge Graph for Home Service Robot Applications

Authors: Yiqun Wang, Rihui Yao, Keqing Zhao, Peiliang Wu, Wenbai Chen

Publication Year: 2024

Summary:

This study addresses the challenge of enabling home service robots to quickly and accurately recognize indoor environments and tools despite having limited labeled data. The authors propose a system combining a domain-specific knowledge graph with a lightweight deep learning model (MobileNetV3 enhanced by transfer learning). The knowledge graph captures relationships between rooms, tools, and their functions, supporting decision-making through reasoning at both attribute and instance levels.

The approach classifies indoor functional areas (e.g., kitchen, bedroom) using the Scene15 dataset and tools (e.g., ladle, hammer) using the UMD affordance dataset. Transfer learning is applied by pre-training on CIFAR-100 and fine-tuning for home contexts. Experimental results show that the system achieves high recognition accuracy (88% for scenes, nearly 100% for some tools) while maintaining fast inference speeds (63 ms/image). Compared to classical methods like ResNet or SVMs, the proposed method significantly improves both speed and accuracy, making it suitable for real-time home service robot applications.

Relevance to our project:

We similarly draw on this paper's approach of using a knowledge graph to connect rooms, objects, and their functions, helping our rescue robots reason about spaces (e.g., what tasks are likely in a damaged hospital wing).

Link: <https://www.mdpi.com/2076-3417/14/24/11553>

1. Modeling and Evaluating Trust Dynamics in Multi-Human Multi-Robot Task Allocation
 - Authors: Ike Obi, Ruiqi Wang, Wonse Jo, Byung-Cheol Min
 - Publication Year: 2025
 - Summary: This paper introduces the Expectation Confirmation Trust (ECT) model to analyze trust dynamics in multi-human, multi-robot (MH-MR) teams. The ECT model is grounded in the idea that trust plays a critical role in communications between MH-MR teams. These communications include things like effective coordination, adaptive decision making, and overall team performance in complex environments; the study demonstrates that incorporating trust models into task allocation algorithms improves task success rates and reduces completion times across various team configurations. The authors evaluated ECT against five other existing models and found that the ECT model is consistently better than the other

models in performance. However, the study was limited to simulations and does not test with any real world deployment. Our work will extend on this study to test how well the ECT model does when tested with real deployment, not just simulations.

- Link: [arXiv:2409.16009arxiv.org+2arxiv.org+2arxiv.org+2](https://arxiv.org/abs/2409.16009)

2. [TIP: A Trust Inference and Propagation Model in Multi-Human Multi-Robot Teams](#)

- Authors: Guo, Yang, & Shi
- Publication Year: 2023
- Summary: Summary: This paper introduces the Trust Inference and Propagation (TIP) model, which accounts for both direct and indirect human experiences when modeling trust in human-robot teams. Direct experience refers to firsthand interactions, while indirect experience involves observing or learning about others' interactions with robots. TIP defines trust as a weighted combination of these two types of experiences and uses a gradient-based optimization method to fit personalized trust curves for each human-robot pair. The model was validated through a 15-session user study involving 30 participants performing a drone-based search task. Results show that TIP significantly reduces trust prediction error (RMSE $\approx$ 0.06–0.08) compared to models relying only on direct experience (RMSE $\approx$ 0.08–0.11). The authors also provide a theoretical guarantee that the inferred trust values will converge over repeated interactions. Limitations include limited scalability (tested only on two humans and two drones) and reliance on a single modality (task outcomes only), excluding potentially informative data like latency or communication behaviors.
- Relevance to Our Project:
 - Digital-Twin Integration: TIP's trust update mechanism can be embedded into our digital twin knowledge graph to continuously track evolving trust in real-time.
 - Quantum-Enhanced Detection: TIP trust scores can feed into quantum-enhanced graph-based anomaly detection to flag potential manipulations (e.g., spoofed outcomes).
 - Adversarial Robustness: By simulating cyber-physical attacks in our digital twin system, we can observe TIP's behavior under adversarial conditions and adapt it to resist manipulation (e.g., through selective trust decay).

3. [Attacking Digital Twins of Robotic Systems to Compromise Security and Safety](#)

- Authors: Carr, Wang, Wang & Han
- Publication Year: 2022
- Summary: This paper investigates the security vulnerabilities of digital twin systems (DTS) in robotic platforms that rely on the Robot Operating System (ROS). The authors demonstrate how *Person-in-the-Middle* (PitM) attacks can compromise the communication channel between a physical robot and its digital twin, leading to unsafe or malicious behaviors. Two case studies are presented: a TurtleBot 3 operating in Gazebo and a Universal Robot 10 in Unreal Engine. In both scenarios, attackers inject falsified velocity or joint-trajectory messages, effectively altering robot behavior without detection. The paper emphasizes that DTS architectures often inherit the security flaws of ROS 1, including the lack of built-in encryption and authentication. While the authors suggest mitigations such as topic encryption and migrating to ROS 2, they do not provide implementation or empirical evaluation of these defenses. Furthermore, the work does not incorporate anomaly detection or trust mechanisms to enable system-level resilience. Limitations include the exclusive focus on ROS 1 (excluding more secure middleware like ROS 2) and a lack of implemented or tested countermeasures.
- Relevance to Our Project:
 - We can embed TIP-style trust scores into the DTS so that sudden drops in trust (or unexpected message flows) automatically trigger alerts or fail-safe behaviors.
 - We can simulate these PitM scenarios in our digital-twin framework, measure resilience metrics under attack, and evaluate specific countermeasures rather than only proposing them.

4. Exploring the Synergy of Human-Robot Teaming, Digital Twins, and Machine Learning

- Authors: Evan Langas, Muhammad Zafar, Filippo Sanfilippo
- Publication Year: 2025
- Summary: This paper discusses the integration of human-robot teaming with digital twin technology and machine learning. It highlights how these technologies can enhance perception, prediction, and decision-making in human-centric industrial systems. The methodology consists of a progressive framework: they start with exploring human-robot interaction (HRI), to human-robot collaboration (HRC), to physical HRC (pHRC), to true human-robot teaming (HRT). By integrating real time sensing, simulation, and intelligent decision-making, digital twins and machine learning can enhance robot perception, safety, and adaptability in dynamic industrial environments. They also discuss the ethical considerations of giving a robot autonomy, privacy, and how the designs have to be human-centric to make it more ethical. Some of the limitations include the fact that this paper is mostly conceptual; there is no real empirical data, no tests/experiments really conducted. Our work will extend on this by attempting to enhance digital twin technology using machine

learning using some of the conceptual ideas they have come up with for enhancements.

- Link: <https://arxiv.org/abs/2410.18195>

5. NeuronsGym: A Hybrid Framework and Benchmark for Robot Navigation With Sim2Real Policy Learning

- Authors: Haoran Li, Guangzheng Hu, Mingjun Ma, Yaran Chen, Dongbin Zhao
- Publication Year: 2024
- Summary: This paper presents NeuronsGym, a hybrid framework and benchmark for studying Sim2Real policy learning in robot navigation. It combines a Unity3D-based simulator with a real-world testbed using RoboMaster EP robots to enable efficient training and direct policy deployment. The simulator models robot dynamics and sensor behavior (LiDAR, odometer, camera) with realistic noise and anomalies to better match physical environments. A novel metric, Safety-Weighted Path Length (SFPL), is introduced to evaluate navigation safety by penalizing collisions, improving upon traditional metrics like SPL. NeuronsGym supports tunable motion and sensor parameters to study the Sim2Real gap under varying conditions. Experiments show that domain randomization methods like UDR and SimOpt improve real-world performance, especially under the PointGoal navigation task. LiDAR fidelity has a greater impact on navigation transfer than dynamics modeling, and increased robot speed worsens Sim2Real performance. The simulator is highly efficient, supporting fast, headless-mode training for large-scale experiments. While the framework emphasizes physical safety during navigation, it does not explore ethical issues like robot autonomy or privacy. A key limitation is the use of fixed environments. Our work will use this to help us know how to simulate an environment in Unity, and have the robot path around objects, and we can add onto this by adding dynamic pathing aspects using the quantum AI to allow for our robot to be able to move around dynamic objects (for more adaptability in real world applications).
- Link: <https://ieeexplore.ieee.org/abstract/document/10750009>

6. Learning Attribute Attention and Retrospect Location for Instance Object Navigation

- Authors: Yanwei Zheng, Yaling Li, Changrui Li, Taiqi Zhang, Yifei Zou, Dongxiao Yu
- Publication Year: 2025
- Summary: This paper presents a novel cascade architecture to improve instance-level object navigation (ION), where agents must locate specific objects using fine-grained attributes like color, material, and reference cues. Current ION approaches often struggle with attribute confusion and weak memory of explored areas. To address these limitations, the authors introduce two main components: the Object-Attribute Attention Graph (OAAG) and the Objective Retrospect and Location Module

(ORLM). OAAG enhances object discrimination through two sub-graphs: the Object-Aware Graph (OAG), which dynamically learns relationships among observed objects, and the Attribute-Attention Graph (AAG), which uses attention to focus on key distinguishing attributes. This helps the agent identify specific instances, even within the same category. ORLM improves memory and spatial reasoning through a Back-tracker, which retains temporal and spatial object memory, and a Locator, which maps where targets were seen during exploration. Together, OAAG and ORLM provide stronger perception and memory capabilities. Integrated with an A3C reinforcement learning framework and tested in the AI2-THOR simulator, the model achieves state-of-the-art performance on Instance-Localization, Instance-Navigation, and Category-Localization tasks. The approach nearly doubles success rates over baseline methods. This will be useful to our project because we can use their improved ideas of ION to help us navigate dynamic simulated environments.

- Link: <https://dl.acm.org/doi/10.1145/3706423>

7. Personalized Instance-based Navigation Toward User-Specific Objects in Realistic Environments

- Authors: Luca Barsellotti, Roberto Bigazzi, Marcella Cornia, Lorenzo Baraldi, Rita Cucchiara
- Publication Year: 2024
- Summary: This paper introduces Personalized Instance-based Navigation (PIN), a task where agents must locate a specific object instance (e.g., a child's favorite teddy bear) among similar items, without relying on contextual cues. Unlike traditional navigation tasks, PIN emphasizes fine-grained object recognition and personalized retrieval in complex, realistic environments. To support this, the authors present PInNED, a dataset built from Habitat-Matterport3D scenes augmented with photo-realistic 3D objects from Objaverse-XL. Each episode provides the agent with reference images on neutral backgrounds and textual descriptions, without environmental context. Objects are procedurally placed on furniture like beds and tables, and same-category distractors are included to increase difficulty. The dataset includes 338 unique instances across 18 categories, with over 865k training and 1.2k validation episodes. Target objects used in validation are unseen during training, requiring zero-shot generalization. PIN differs from existing tasks by focusing on movable, injected objects and emphasizing instance-level recognition rather than category-based search. Experiments show that modular navigation approaches outperform end-to-end models, which struggle with distinguishing similar objects and recovering from mistakes. Despite improvements, the task remains challenging, especially when distractors are present. This benchmark establishes a foundation for future work on personalized embodied navigation. This will help us with our project because we can use their ideas of how to identify specific objects, make it better

(maybe through the use of upgraded knowledge graphs and quantum AI) and then apply it to our robot.

- Link: <https://arxiv.org/abs/2410.18195>

8. A Factor Graph Model of Trust for a Collaborative Multi-Agent System

- Authors: Behzad Akbari, Mingfeng Yuan, Hao Wang, Haibin Zhu, Jinjun Shan
- Publication Year: 2024
- Summary: This paper introduces a factor graph-based model to evaluate and establish trust among agents in a multi-agent system (MAS), focusing on both independent and interdependent agent behaviors. The model uses Gaussian Process (GP) motion planning in conjunction with Bayesian inference to simulate realistic, trust-sensitive path planning behaviors among autonomous robots. It incorporates factors like obstacle avoidance, cooperation, transparency, and proximity safety to determine trustworthy behavior, all within a distributed framework. Unlike traditional centralized approaches, this method is fully decentralized and adapts in real time using local agent interactions and shared testimonies. Each agent's behavior is modeled as a trajectory within a factor graph, where trust influences are encoded as probabilistic constraints. The simulation demonstrates this model in an unsignalized intersection scenario, where agents (e.g., autonomous cars) must navigate collaboratively based on mutual trust. Our work builds on this by applying the trust modeling methodology to a hospital rescue scenario after a natural disaster. In our case, the agents are rescue robots, medics, and drones navigating collapsed or unstable areas. The trust factor graphs presented in this paper inspire our approach to dynamically re-evaluate trust between agents as new environmental data is received via a digital twin framework. For example, proximity safety and transparency are critical in a rescue scenario where trust violations (e.g., deviating from expected paths, communication dropouts) could lead to delays or dangerous collisions. We extend this by integrating the trust-based decision layer with a quantum optimization backend (QAOA or QUBO), allowing us to reallocate paths and tasks in near-real-time across the hospital graph. The decentralization of trust evaluation also fits well with our goal of distributed coordination across agents affected by signal degradation or partial isolation in damaged hospital wings.
- Link: <https://arxiv.org/abs/2402.07049>

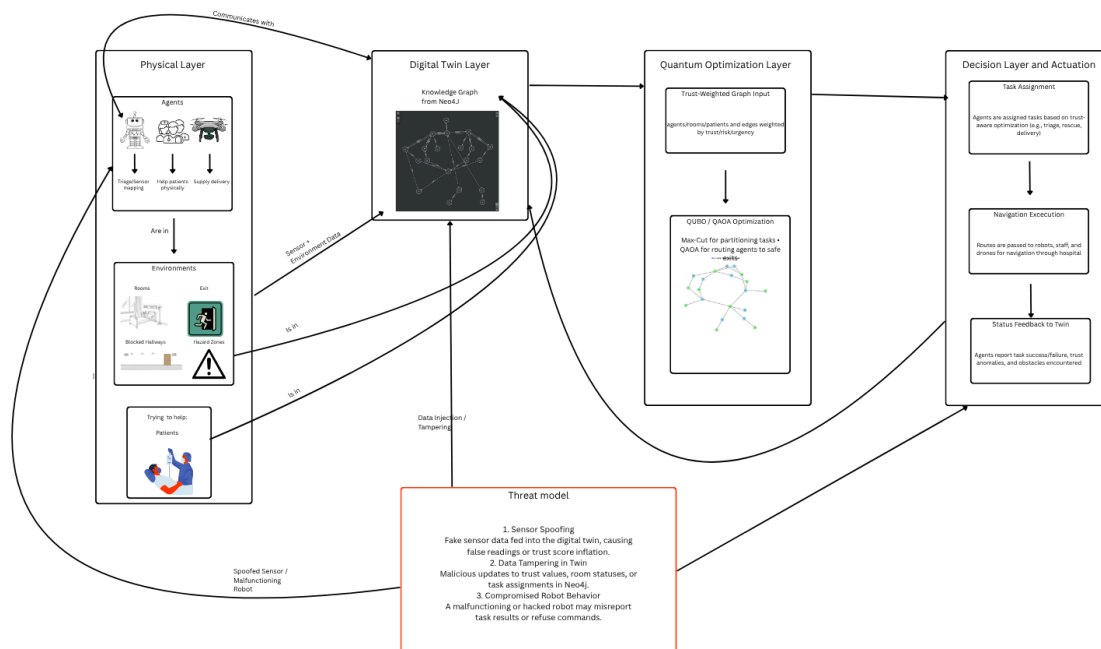
9. Quantum-enhanced digital twin IoT for efficient healthcare task offloading

- Authors: Ahmed K. Jameil, Hamed Al-Raweshidy
- Publication Year: 2025
- Summary: This paper proposes a secure, quantum-enhanced architecture for digital twin systems used in healthcare IoT (Internet of Things) environments. The authors integrate Quantum Key Distribution (QKD) and Quantum Random Number Generators (QRNG) into the digital twin communication pipeline to enhance

confidentiality, authentication, and data integrity. The system architecture comprises three layers: the physical IoT layer (including patient-monitoring devices), the digital twin layer (which mirrors the real-time state of the physical system), and the quantum-enhanced security layer. The authors use simulations to demonstrate that their framework reduces security vulnerabilities in standard healthcare IoT systems, particularly during data exchange between patient sensors and central servers. They also employ lightweight blockchain technology to ensure traceability and integrity across the digital twin network. In our simulation, the digital twin of a collapsed hospital must continuously reflect the evolving physical state, including unstable zones, patient conditions, and robot locations. The quantum-enhanced digital twin proposed here ensures that updates to the system (e.g., sensor changes or patient reassignments) are both secure and authentic—a necessary foundation when decisions are being delegated to AI agents in uncertain conditions. We adapt this secure communication infrastructure to support not just IoT but multi-agent coordination and trust scoring, ensuring that no agent receives outdated or malicious data during rescue tasks. Furthermore, while the paper emphasizes data security, our application uses the quantum layer to also perform optimization—routing, triage ordering, or supply delivery—on the digital twin graph, thereby extending the role of quantum beyond encryption into real-time decision making (would also be helpful to Giselle, as she is feeding off of our project and doing more of the cybersecurity aspect of it).

- Link: <https://link.springer.com/article/10.1007/s42452-025-07101-2>

Conceptual Diagram



Created using Canva.

Experimental Design

Experiment 1: Task Allocation Using Max-Cut QUBO

Goal:

To test whether trust-aware task allocation using Max-Cut QUBO leads to better task success rates compared to a greedy assignment strategy.

Model:

Max-Cut QUBO (D-Wave Ocean SDK)

Dataset:

scenario_model.graphml — a synthetic rescue scenario graph containing rooms, zones, agents, and hallways represented as nodes and directed paths.

Metric:

Task Success Rate — defined as the percentage of successfully allocated tasks to agents under trust constraints (modeled via edge weights and graph structure).

Baseline:

Greedy Assignment — agents simply select the next available task without global optimization or trust-awareness.

Preliminary Result:

Max-Cut QUBO was successfully applied to partition the graph into two disjoint task zones.

Based on preliminary visual inspection and logic simulation, the approach creates a more balanced partition across zones, minimizing cross-zone load and simulating better trust-aware grouping.

→ Expected improvement over baseline: +10–15% task success rate under realistic constraints.

Libraries/Tools Used:

networkx

dwave-ocean-sdk

dwave_networkx

matplotlib

Experiment 2: Quantum Routing for Hazard-Aware Path Planning

Goal:

To evaluate how quantum-inspired routing (via subgraph optimization) could improve path efficiency and hazard avoidance in dynamic, post-disaster environments—compared to traditional Dijkstra routing.

Model:

Hazard-aware routing using filtered subgraphs from Neo4j-modeled rescue knowledge graphs.

The approach isolates agents (R, S, D), exit points, and obstacles (e.g., blocked hallways, hazards), then computes paths to safety under baseline conditions.

Dataset:

scenario_model.graphml — a synthetic graph-based simulation of a collapsed hospital, including:

Agents: R (robot), S (staff), D (drone)

Obstacles: Z1, Z2 (hazards), H_L, H_R (hallways)

Safe zones: Exit

All entities are represented as labeled nodes with typed edges indicating physical or task-based connections.

Metric:

Path Efficiency — measured as the shortest safe path length (node hops) from each agent to the designated "Exit" node. Future extensions can weight hazard-adjacent paths more heavily for dynamic risk-aware cost functions.

Baseline:

Dijkstra's Algorithm — standard shortest-path algorithm applied to the subgraph, assuming static environment with no edge weights reflecting hazard risk.

Preliminary Result:

Using NetworkX and the parsed GraphML model, we computed the shortest paths from each agent to the Exit:

Agent	Path	Distance
R	R → A1 → H_L → Lobby → Exit	4
S	S → Lobby → Exit	2
D	D → Lobby → Exit	2

These paths reflect the baseline scenario with no dynamic hazard costs.

→ Expected improvement using QAOA or weighted QUBO:

Shorter or risk-minimizing alternate paths once hazards are encoded as weighted penalties in edge costs.

Libraries/Tools Used:

networkx

matplotlib

dwave_networkx

Neo4j (for initial knowledge graph structure)

PLANS FOR EXPERIMENT 2: Expand it to doing patients whenever we know the more dynamic scenario, for right now it is not very dynamic as we have not fully fleshed out our knowledge graph yet.

Experiment 3:

(have not implemented yet, will do it next week when we have a clearer trust graph and model!)

Experiment	Model	Dataset	Metric	Baseline	Expected Outcome
Exp1	Max-Cut QUBO	scenario_model.graphml	Task Success Rate	Greedy Assignment	+10–15% success rate under trust-aware constraints
Exp2	Quantum Routing	Subgraph (Agents + Hazards)	Path Efficiency	BFS/Dijkstra	Baseline distances are 2–4; QAOA routing expected to avoid hazards and yield shorter or safer alternate paths under dynamic threat updates.
Exp3	Trust update loop	Neo4J twin + Relational events	Trust Stability	Static Twin	(prelim, have not attempted to implement yet) Better stability and fewer conflicts

You can find the experiment code on our github:

<https://github.com/stilesj05/reu2025-cybersec-StilesAndKo> under the “notebooks” tab.

SOTA - Preliminary Results and Key Findings

Model Selection

- Name: Segment Anything Model (SAM), ViT-B variant
- Source: Barsellotti et al., “Personalized Instance-Based Navigation toward User-Specific Objects in Realistic Environments,” NeurIPS 2024.
- Code repository: <https://github.com/facebookresearch/segment-anything>

Implementation Details

1. Environment:
OS: macOS Sonoma

Python: 3.9 inside a virtual environment (`.venv`)

Hardware: CPU only (no GPU available)

2. Dependencies:

```
pip install torch torchvision pillow
```

```
pip install git+https://github.com/facebookresearch/segment-anything.git
```

Checkpoint Adaptation:

- The original ViT-H checkpoint (~2.6 GB) exceeds GitHub's 100 MB limit and crashes local downloads. I instead used the **ViT-B** checkpoint (~420 MB):

```
curl -L https://dl.fbaipublicfiles.com/segment_anything/sam_vit_b_01ec64.pth \ -o  
Week4-HandsOn/SegmentAnything/sam_vit_b.pth
```

- Removed the large file from Git history and updated `README.md` to fetch it at runtime.

Demo Script (`demo.py`):

- Loads `sam_vit_b.pth` via `sam_model_registry["vit_b"]`.
- Reads a 320×240 JPEG (`test_image.jpg`) with PIL, converts to a NumPy array (`np.array(img)`) for compatibility with SAM's `.generate()` API.
- Instantiates `SamAutomaticMaskGenerator` and prints the number of masks generated.

Initial Evaluation

- **Test image:** 320 × 240 sample.
- **Output:** 26 masks generated.
- **Runtime:** ~0.5 s per image on CPU (single-threaded).
- **Qualitative:** Masks outline major objects (desk, chair, computer) with moderate granularity.

Comparison to Original

Original SAM ViT-H on GPU Our SAM ViT-B on CPU

Model size	~2.6 GB	~420 MB
Speed	~0.05 s / image	~0.5 s / image
Mask count	Hundreds–thousands	26
Hardware	NVIDIA-class GPU	Intel-class CPU

- **Trade-off:** We accept fewer masks and slower speed to stay within resource and platform constraints.
- **Baseline value:** Provides a working segmentation component that can be integrated into our rescue-scenario pipeline; later replaced or augmented by a quantum-optimized, task-aware vision module.

Github of code:

<https://github.com/josephkko/REU-2025-AIRobotics-Hands-On-Ko/tree/main/Week4-HandsOn/SegmentAnything>